CAPTIVEAIRE

One System. One Source.



SUBMITTAL

PROJECT NAME

2102482 - McComb, MS

JOB NUMBER

8252713

LOCATION

MCCOMB, MS 39648

CUSTOMER APPROVAL TO MANUFACTURE

- ☐ APPROVED AS NOTED
- ☐ APPROVED WITH NO EXCEPTIONS TAKEN
- ☐ REVISE AND RESUBMIT

NOTES:

- CaptiveAire to provide SA and Space Temperature and Humidity Sensors.
- Provide BACNet Gateway (BACNet IP)
- Supply Air Temperature Reset based on OAT to be provided by BMS contractor.
- Airflow monitoring -read by BMS
- BMS interface needs to be included
-Lead time 3-4 weeks
-Schedule on SATR to be provided by DG at a later time

DG Architectural and Engineering Facility Solutions, PLLC

- ☐ No Exceptions Taken
- ☐ Revise and Resubmit
- ☑ Exceptions Taken as Noted
- ☐ Rejected

November 17th, 2025

Reviewed By CaceresW

Review is only for general conformance with the design concept of the project and compliance with the information given in the construction documents. The contractor is responsible for dimensions to be confirmed and coordinated at the job site, for information that pertains to the fabrication process and to the means, methods, techniques, sequences and procedures of construction. Review does not authorize changes to contract sum, time or completion dates.

DATE OF SUBMITTAL

10/14/2025

SUBMITTED TO

GEORGE HOLDEN & A

1395 S. MARIETTA PKWY, BLDG 100, STE 105, MARIETTA, GA 30067 PHONE (800) 291-2452 EMAIL REG122@CAPTIVEAIRE.COM

Job Name: 2102482 - McComb, MS

Date: October 14, 2025 1:33 PM

CAS-HVAC1-I.125-13-10T-DOAS (#1) DOAS Weight: 1450 lbs.

RTU with Indirect Fired Heat and 13" Direct Drive Plenum Fan, 1 Furnace, Electronic Full Modulation, Constant 81% Efficiency, and 6:1 Max Turndown for NG, (5:1 Max Turndown for LP). Stainless Steel Burner and Heat Exchanger.

SUPPLY MOTOR

Model DTP0014 • 1.000 HP, 3 Phs, 460 V, 60 Hz, 1.4 FLA, ODP, Premium (E-Plus3) Eff.

SUPPLY PERFORMANCE

Supply Air:	1000 CFM	RPM:	1381
Return Air:	0 CFM	Outdoor Air:	1000 CFM
Tip Speed:	4700 ft/min	BHP:	0.5360
SP:	0.942" w.g. (0.750" Ductwork Discharge ESP + 0.192" Opt.)		
Altitude:	413 ft		

UNIT SHOULD BE ABLE TO HANDLE ESP RANGE FROM 0.25" TO 1"

BLOWER

13 Inch Plenum Fan for size 1 RTU. Hub specified to match motor shaft.

TEMPERATURE CONTROL

MANUFACTURING FACILITY • Occupied/Unoccupied Scheduling • Space Temp Control Heating and Cooling Activation Based On Either Set Point • Manual/Auto Blower Mode Smoke Detector Interlock • Freezestat • Space Sensor Ships Loose. MUA

AIRFLOW DIRECTION

- Side Discharge - Air Flow Left -> Right

SUPPLY INSTALLATION INFORMATION

Gas Pressure:	1 lb. - 5 lb.
Insurance Requirements:	No Insurance Requirement (ANSI)
Unit Voltage:	3 phs 460 V 60Hz
Unit Main Input:	29.3 Amps MCA • 30 Amps MOP • 460 V • 10 AWG Wire Min.
SCCR:	10 kAmp
A2L Min. Room Area / Airflow / Height:	308.4 ft2 • 555 CFM • 7.2 ft **

*** Minimum room area assumed 7.2' supply diffuser height and is calculated per UL60335-2-40 4th ed. Values based on factory charge. Actual site charge may differ.*

HEATING SCHEDULE

Altitude:	413 ft
GAS HEAT	
Output BTU:	86,181 BTU/Hr
Input BTU:	106,396 BTU/Hr
Based Off:	BTUs Actual Air Density
Gas Type:	Natural
Winter Outdoor Air Dry Bulb Temp:	17.0°F
Temp Rise:	73.0°F
Discharge Dry Bulb Temp:	90.0°F
HEAT PUMP	
Entering Temp:	45.0°F
Max Temp Rise:	48.0°F
Discharge Dry Bulb Temp:	93.0°F
COP:	2.9
Capacity:	48.9 MBH

USING 0.4% DESIGN ENTHALPY? OK IF YES

COOLING SCHEDULE

Outdoor Air Dry Bulb Temp:	100.5°F
Outdoor Air Wet Bulb Temp:	84.0°F
Max Outdoor Air %:	100%
Mixed Air Dry Bulb Temp:	100.5°F
Mixed Air Wet Bulb Temp:	84.0°F
Mixed Air Dew Point:	79.1°F
Leaving Dry Bulb Temp:	51.9°F
Leaving Wet Bulb Temp:	51.7°F
Leaving Dew Point:	51.6°F
Moisture Removal Rate:	55.4 Lb/Hr
Total Capacity:	118.3 MBH
Sensible Capacity:	52.8 MBH
Latent Capacity:	65.5 MBH
Reheat Coil Capacity:	71.9 MBH
Reheat Coil Leaving Dry Bulb Temp:	70.0°F
Reheat Coil Leaving Wet Bulb Temp:	59.0°F
Reheat Coil Leaving Relative Humidity:	52%
IEER:	14.2
ISMRE2:	6.2

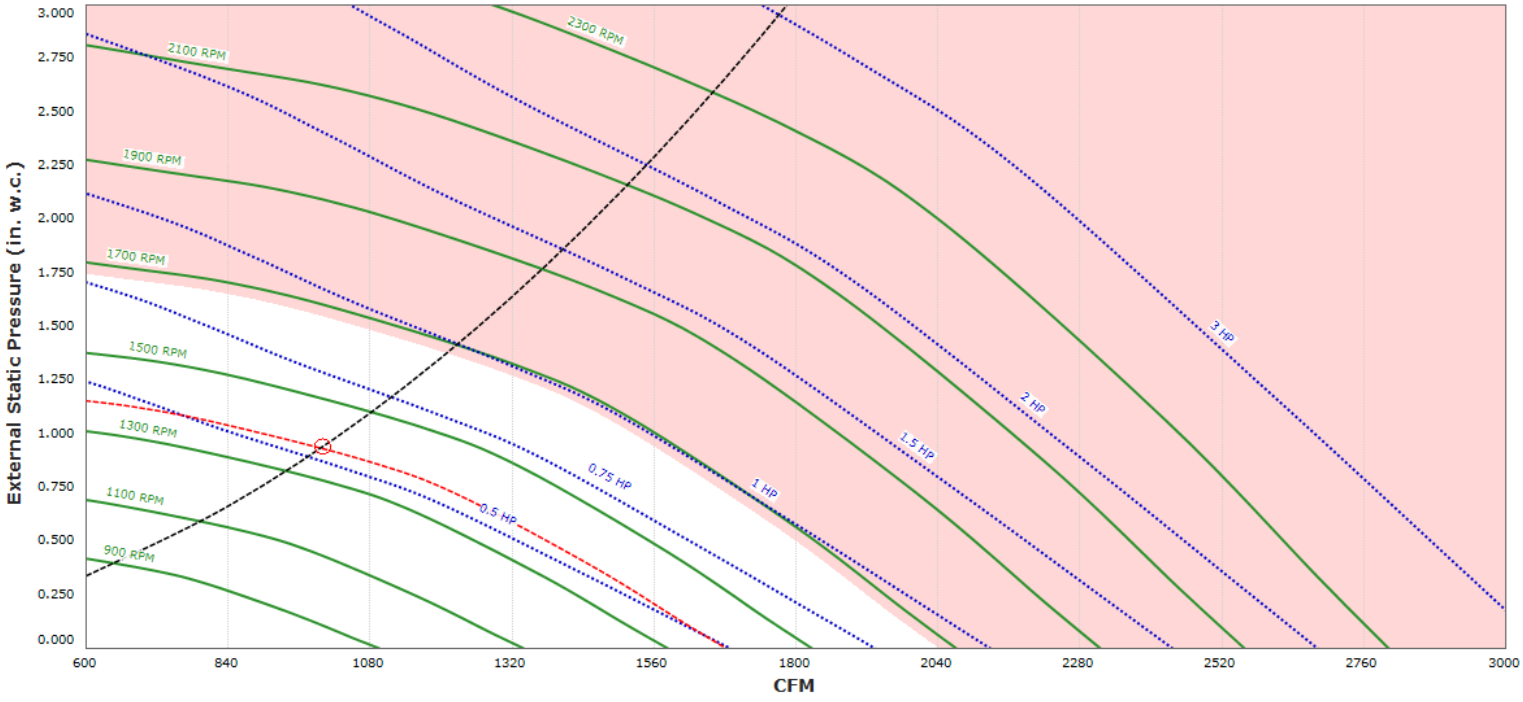
CONFIGURED OPTIONS

COORDINATE WITH SITE REQUIREMENTS

- Gas Pressure Gauge, 0-35", 2.5" Diameter, 1/4" Thread Size
- Gas Pressure Gauge, 0 to +10 Inches Wc., 2.5" Diameter, 1/8" Thread Size, Rear Thread
- Cooling Override
- Single Point Electrical Connection for RTU. QNTY 1 750va Transformer Used. If a Non-DCV Prewire controls this unit, the #28, #47, "MA", or "E2" Option Prewire must be selected. Do not provide supply starter in prewire.
- Unit will be shipped with construction mode ON. This is used to provide tempering for a building in construction. Activation is set to Intake, Tempering is set to Discharge, Mixing Box is set to 100% OA and HMI number is 1.
- RTU Blower Door Switch. Used to turn on DOOR INTERLOCK Schematic Layer. Does Not Include Door Switch.
- 2" Merv 13 Filters for Size 1 RTU. QTY 4, 16 x 16 x 2 Merv 13 Filters
- 2" Merv 8 Filters for Size 1 RTU. QTY 4, 16 x 16 x 2 Merv 8 Filters
- Overheat Stat factory set at 80°F and 10 minutes. Prevents unit from overheating the space when cooling.
- CFM monitoring for MUA units. Uses rivet nuts, 1/4" airflow tubing and push to connect fittings.
- VFD factory mounted and wired in unit control vestibule.
- RTU Fixed 100% Outdoor Air Intake.
- Gas regulator - 1/2" diameter thread size, 40 PSI Inlet Gas Pressure. 600 CFH, 6" - 14" Outlet Pressure. Includes 071-03409-002 "Green" spring and 5/16" 40 PSI orifice (018-01029-035).
- RTU Size 1 Side Discharge
- Occupied Scheduling Defaulted for IBT/RTU.
- Intake Firestat. Set to 135°F
- Freezestat
- Discharge Firestat. Set to 240°F
- CASLink building monitoring system communications module. Requires internet & field wired ethernet connection or 3G cellular service. Includes Rev 3 Comm Module, RJ45 to modbus converter, 3 FT cat5 cable, and 1 FT of shielded twisted pair.

PROVIDE BMS INTERFACE

- Commercial Smoke Detector Interlock (Detector By Others)
- Clogged Filter Switch with Notification on HMI
- GFCI 15 amp Powered Convenience Outlet For RTU1 Enclosure. Includes receptacle, jbox, cover and transformer.
- RTU Size 1 Hail Guard for outdoor coil. Includes magnets and washers for installation.
- RTU Size 1 No Return
- Option for Metal Mesh Filters in RTU1 OA Intakes
- VAV (Variable-Air-Volume) Wiring Package for Commercial Fans. Manual Speed Control Variable Frequency Drive Included
- Supply Variable Frequency Drive - 1 HP Max., 400/480 V, Three Phase, 2.4/2.1 A Max., NEMA 1 Enclosure, with 2RJ-45 FOR MODBUS
- Mount Load Reactor in Fan.
- 10 Ton Modulating Cooling Option with Heat Pump, 460/480V. Size 1 Cabinet, R454B Refrigerant, Variable Speed Inverter Duty Compressor, ECM Condensing Fan(s).
- RTU Compressor Oil Sensor Factory Installed. M12 Style Connector.
- Low ambient cooling controls down to 0F ambient temperatures. To be used in recirculating applications.
- R454B Refrigerant Leak Detector Option. Includes 20% LFL sensor and harness.
- 10 Ton Heat Pump Modulating Reheat Option for Space Control - Size 1 Cabinet, R454B
- Remote MSTP BACnet monitoring
- Option to factory install RTU compressor sound blanket. Used on 10T and 12.5T units all voltages.
- Base Mount Vibration Isolators. Option for the size 1 RTU. Max Weight = 450 lbs. 1.31" Deflection. 1/2" Bolt Diameter. Set of 4. Red Spring.
- Blower Delay Stop. Allows supply fan to run after the burner has shut off Factory set at 90 secs for indirect 30 secs for direct.



CAS-HVAC1-I.125-13-10T-DOAS SOUND (1381 RPM) AT 5 FT. IN OCTAVES:

1	2	3	4	5	6	7	8
84.8	81.1	77.2	74.2	74.3	68.9	64.5	64.2

LWA:	78.2	Sones:	16.3	DBA:	67.5

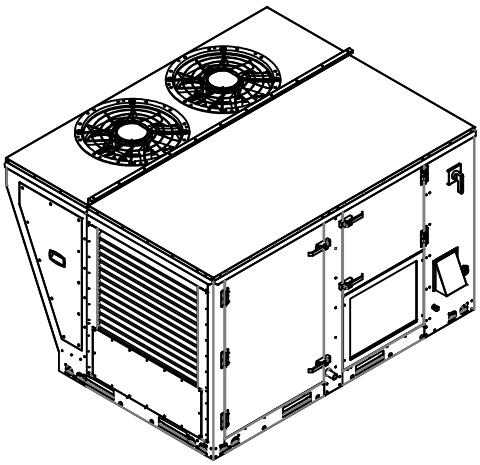
Note: Sound data across operational range. Tested in accordance to AHRI Standard 270/370.

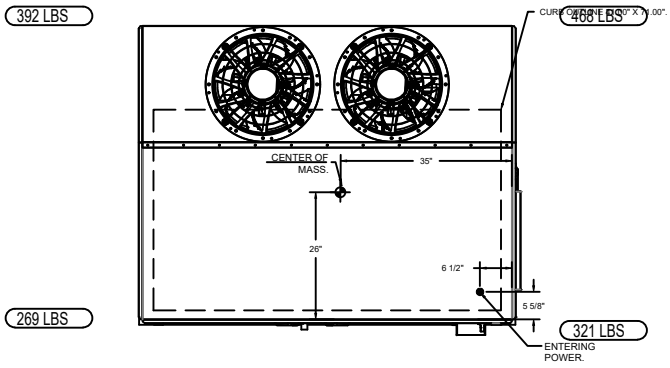
FAN #1 CAS-HVAC1-I.125-13-10T-DOAS - HEATER (DOAS)

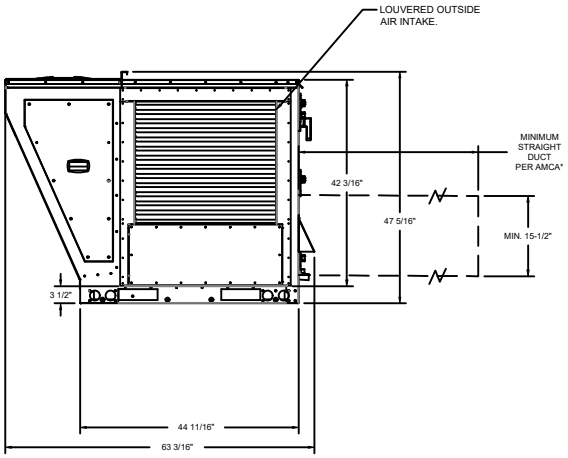
NOTES:
1. DO NOT OBSTRUCT OUTSIDE AIR INLET, OUTSIDE AIR COIL OR OUTSIDE AIR FAN.
2. ( ) DENOTES CORNER WEIGHT.
3. CONNECTION FROM BREAKER TO UNITS SAFETY DISCONNECT SWITCH TO BE COPPER WIRE ONLY.
4. EXTERIOR GAS CONNECTION PROVIDED BY FACTORY WITH QUICK SEAL AND ANTI-ROTATION BRACKET.

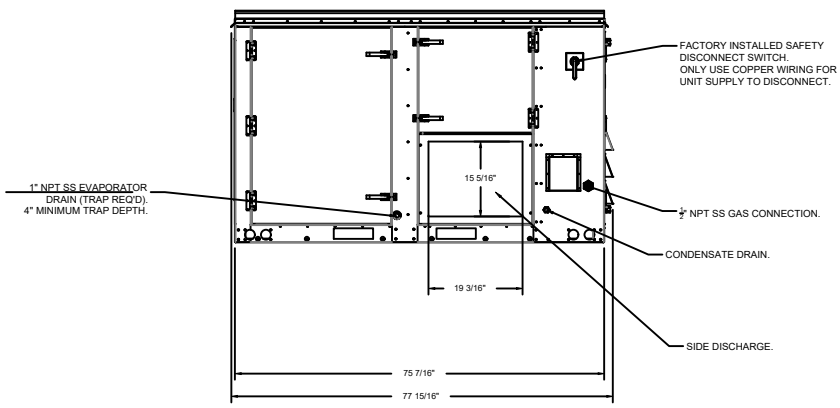
*NOTE: INTEGRAL CO2 MONITORING AND CONTROL CAPABILITIES FOR ALL SPACE MOUNTED THERMOSTATS.

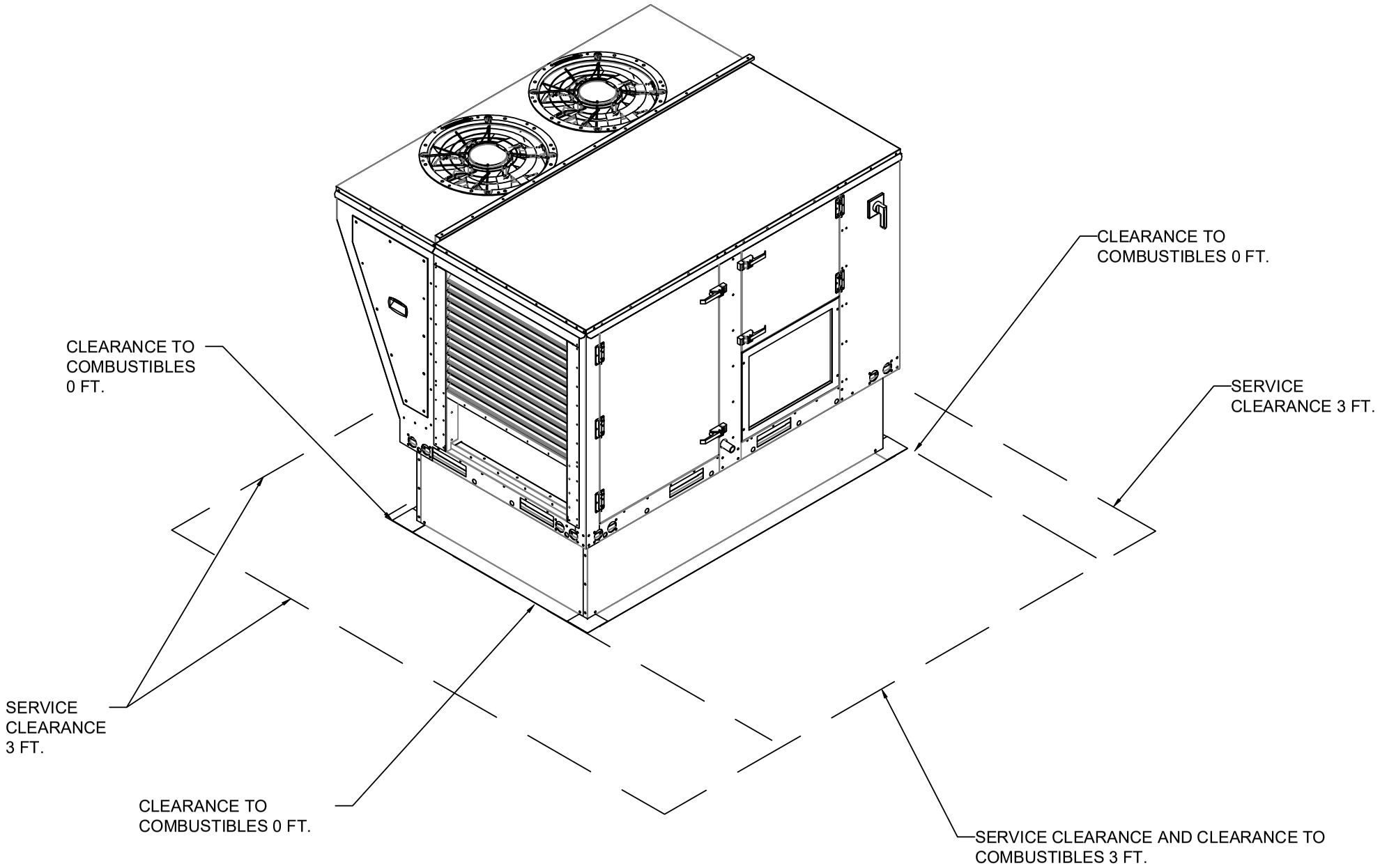
*NOTE: SUPPLY DUCT MUST BE INSTALLED TO MEET SMACNA STANDARDS. A MINIMUM STRAIGHT DUCT LENGTH MUST BE MAINTAINED DOWNSTREAM OF UNIT DISCHARGE AS OUTLINED IN AMCA PUBLICATION 201. WHEN USING RECTANGULAR DUCTWORK, ELBOWS MUST BE RADIUS THROAT, RADIUS BACK WITH TURNING VANES. FLEXIBLE DUCTWORK AND SQUARE THROAT/SQUARE BACK ELBOWS SHOULD NOT BE USED. ANY TRANSITION AND/OR TURNS IN THE DUCTWORK WILL CAUSE SYSTEM EFFECT. SYSTEM EFFECT WILL DRASTICALLY INCREASE STATIC PRESSURE AND REDUCE AIRFLOW. DO NOT RELY ON UNIT TO SUPPORT DUCT IN ANY WAY. FAILURE TO PROPERLY SIZE DUCTWORK MAY CAUSE SYSTEM EFFECTS AND REDUCE PERFORMANCE OF THE EQUIPMENT.
SUGGESTED SUPPLY STRAIGHT DUCT SIZE IS 15-1/2" x 19-1/4".



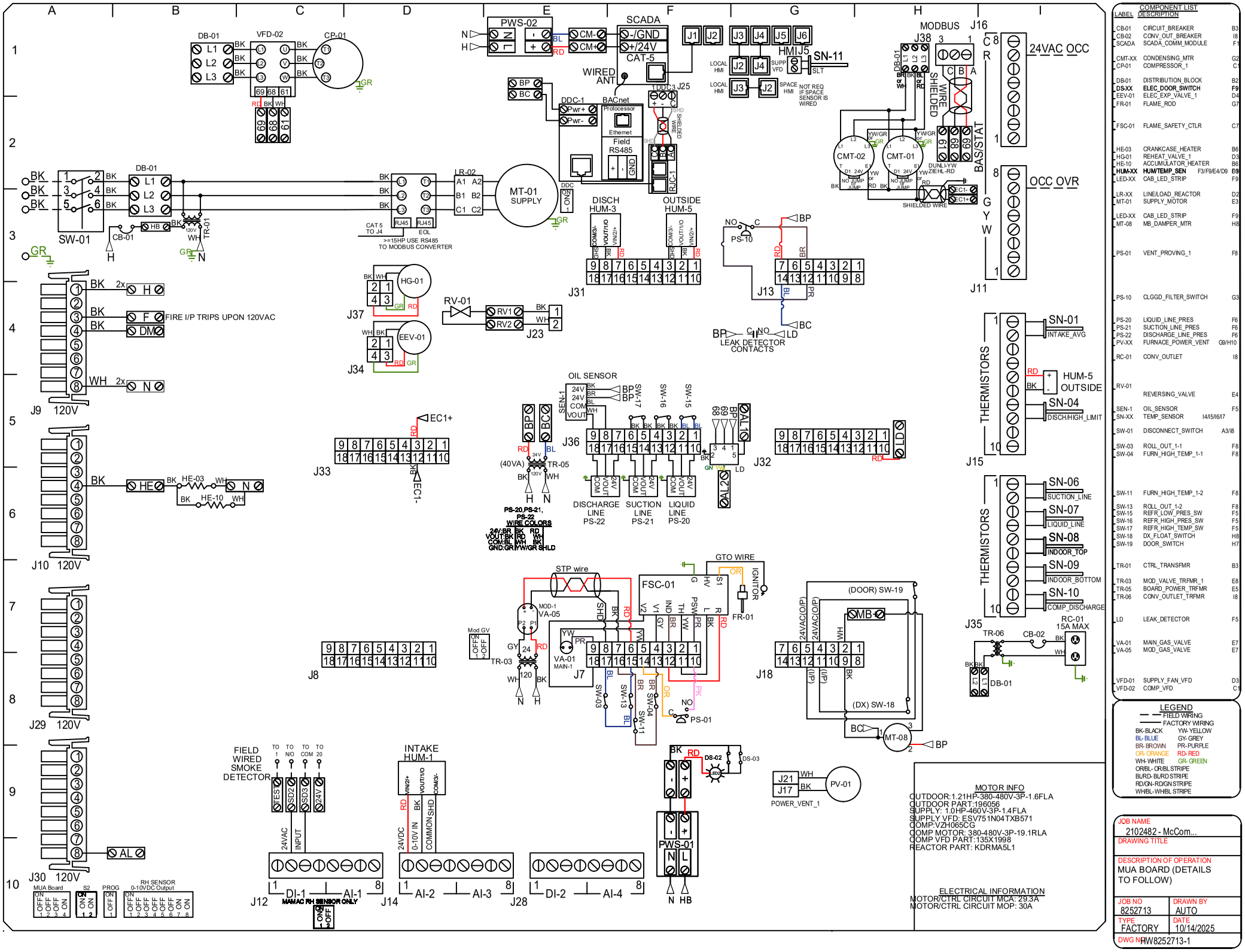








SIZE 1 - 2 CONDENSERS



COMPONENT LIST

LABEL	DESCRIPTION	
CB-01	CIRCUIT_BREAKER	B3
CB-02	CONV_OUT_BREAKER	I8
SCADA	SCADA_COMM_MODULE	F5
CMT-XX	CONDENSING_MTR	G2
CP-01	COMPRESSOR_1	C1
DB-01	DISTRIBUTION_BLOCK	B2
DS-XX	ELEC_DOOR_SWITCH	F9
EEV-01	ELEC_EXP_VALVE_1	D4
FR-01	FLAME_ROD	G7
FSC-01	FLAME_SAFETY_CTLR	C7
HE-03	CRANKCASE_HEATER	B6
HG-01	REHEAT_VALVE_1	D3
HE-10	ACCUMULATOR_HEATER	B6
HUM-XX	HUM/TEMP_SEN	F3/F9/E4/D9 D9
LED-XX	CAB_LED_STRIP	F9
LR-XX	LINE/LOAD_REACTOR	D2
MT-01	SUPPLY_MOTOR	E3
LED-XX	CAB_LED_STRIP	F9
MT-08	MB_DAMPER_MTR	H8
PS-01	VENT_PROVING_1	F8
PS-10	CLGGD_FILTER_SWITCH	G3
PS-20	LIQUID_LINE_PRES	F6
PS-21	SUCTION_LINE_PRES	F6
PS-22	DISCHARGE_LINE_PRES	F6
PV-XX	FURNACE_POWER_VENT	G9/H10
RC-01	CONV_OUTLET	I8
RV-01	REVERSING_VALVE	E4
SEN-1	OIL_SENSOR	F5
SN-XX	TEMP_SENSOR	I4/I5/I6/I7
SW-01	DISCONNECT_SWITCH	A3/I8
SW-03	ROLL_OUT_1-1	F8
SW-04	FURN_HIGH_TEMP_1-1	F8
SW-11	FURN_HIGH_TEMP_1-2	F8
SW-13	ROLL_OUT_1-2	F8
SW-15	REFR_LOW_PRES_SW	F5
SW-16	REFR_HIGH_PRES_SW	F5
SW-17	REFR_HIGH_TEMP_SW	F5
SW-18	DX_FLOAT_SWITCH	H8
SW-19	DOOR_SWITCH	H7
TR-01	CTRL_TRANSFMR	B3
TR-03	MOD_VALVE_TRFMR_1	E8
TR-05	BOARD_POWER_TRFMR	E5
TR-06	CONV_OUTLET_TRFMR	I8
LD	LEAK_DETECTOR	F5
VA-01	MAIN_GAS_VALVE	E7
VA-05	MOD_GAS_VALVE	E7
VFD-01	SUPPLY_FAN_VFD	D3
VFD-02	COMP_VFD	C1

LEGEND

FIELD WIRING
FACTORY WIRING

BK-BLACK YW-YELLOW
BL-BLUE GY-GREY
BR-BROWN PR-PURPLE
OR-ORANGE RD-RED
WH-WHITE GR-GREEN
ORBL-OR/BL STRIPE
BLRD-BL/RD STRIPE
RDGN-RD/GN STRIPE
WHBL-WH/BL STRIPE

MOTOR INFO

OUTDOOR:1.21HP-380-480V-3P-1.6FLA
OUTDOOR PART:196056
SUPPLY: 1.0HP-460V-3P-1.4FLA
SUPPLY VFD: ESV751N04TXB571
COMP:VZH065CG
COMP MOTOR: 380-480V-3P-19.1RLA
COMP VFD PART:135X1998
REACTOR PART: KDRMA5L1

ELECTRICAL INFORMATION

MOTOR/CTRL CIRCUIT MCA: 29.3A
MOTOR/CTRL CIRCUIT MOP: 30A

JOB NAME	2102482 - McCom...
DRAWING TITLE	
DESCRIPTION OF OPERATION	MUA BOARD (DETAILS TO FOLLOW)
JOB NO	8252713
DRAWN BY	AUTO
TYPE	FACTORY
DATE	10/14/2025
DWG NO	HW8252713-1

SMV VFD PARAMETER SETTINGS

P100	(START SOURCE) = 01
P150	(TB-30 OUTPUT) = 1
P194	PASSWORD = 225
P410	MODBUS ADDRESS = 21

ADJUST MANUALLY ON ALL DRIVES	
P103	VFD MAX FREQUENCY
P107	00 (IF 120 OR 208 VAC)
	01 IF(230, 480 OR 575VAC)
P108	(MOTOR FLA X 100) / DRIVE OUTPUT RATING
P167	REFERENCE BUILD SHEET
P171	P108 - 1

*NOTE: THE DEFAULT FOR THE DRIVE IS "225".

All external control wires to motor speed control should be 16-20 AWG shielded multi-conductor cables and must not be run in the same conduit or raceway with any high power wiring. Ground Shielded Cable at the drive chassis ONLY.

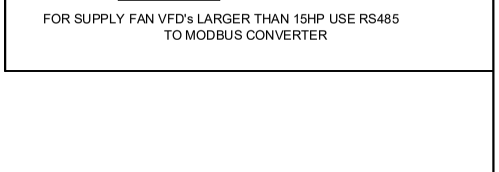
PG. 11, 19, 23 OF THE DRIVE MANUAL DESCRIBES THE PROPER HANDLING OF THE VARIABLE FREQUENCY DRIVE.

IT MAY BE REQUIRED TO FULLY POWER DOWN THE DRIVE AND TURN BACK ON IN ORDER TO INITIATE NEW PARAMETER SETTINGS.

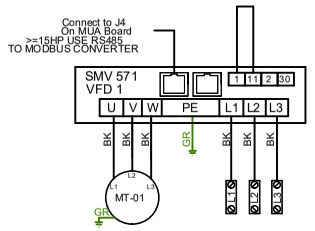
**Min. and Max. Frequency Settings override all other Preset speeds/Parameters. Do not adjust these on the VFD. Min. and Max. Frequency should be adjust on the RTU HMI.

SMV 571 SERIES VFD

Terminal	Description
1	Digital Input:(Start/Stop)
11	Internal DC Supply for External Devices
2	Analog Common
30	Analog Output:Configurable with P150..P155
RJ45	MODBUS COMMUNICATION
PE	Ground Terminal
L1	3 Phase Input or Single Phase Input
L2	3 Phase Input or Single Phase Input
L3 (N)	3 Phase Input (Neutral for 120v)
U	3 Phase AC Motor
V	3 Phase AC Motor
W	3 Phase AC Motor



FOR SUPPLY FAN VFD's LARGER THAN 15HP USE RS485 TO MODBUS CONVERTER



*Note: Not Suitable for Corner Grounded 3 Phase Input power

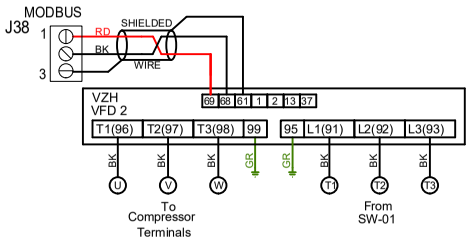
COMPRESSOR DRIVE PARAMETER SETTINGS MUST BE CONFIGURED BEFORE STARTUP

COMPRESSOR VFD 1

0-22	[1610] POWER KW
0-40	[2] PASSWORD
0-60	225
0-61	[1] LCP READ ONLY
0-66	[1] LCP READ ONLY
1-13	VZH065
3-15	[0] NO FUNCTION
3-16	[0] NO FUNCTION
3-41	200.00 s
3-42	200.00 s
3-51	30.00 s
3-52	30.00 s
8-01	[2] CTRL WORD ONLY
8-03	20 SECONDS
8-04	[2] STOP
8-30	[2] MODBUS RTU
8-31	1
8-32	19200
8-33	[0] EVEN PARITY, 1 STOP BIT
8-43-2	1610
8-43-3	1610
8-43-4	1613
8-43-5	**1613**
8-43-6	**1614**
8-43-7	**1614**
8-43-8	1690
8-43-9	**1690**
8-43-10	1634
8-43-12	1694
8-43-13	1694
28-00	DISABLE
28-11	**1440m**
28-12	24h

*Must be programmed using software

*Changing 8-30 will reset 8-01 and 8-03 if programmed after 8-01 and 8-03 are set



VZH SERIES COMPRESSOR VFD

TERMINAL	FUNCTION
13	24+
37	Pressure Switch In
1	Relay 1 to oil solenoid
2	Relay 1 to oil solenoid
61	COM RS-485
68	N RS-485
69	P RS-485
95	INPUT Ground
L1 (91)	3 Phase Input or Single Phase Input
L2 (92)	3 Phase Input or Single Phase Input
L3 (93)	3 Phase Input
T1 (96)	Compressor
T2 (97)	Compressor
T3 (98)	Compressor
99	Compressor Ground

MOTOR INFO
OUTDOOR:1.21HP-380-480V-3P-1.6FLA
OUTDOOR PART:196056
SUPPLY: 1.0HP-460V-3P-1.4FLA
SUPPLY VFD: ESV751N04TXB571
COMP:VZH065CG
COMP MOTOR: 380-480V-3P-19.1RLA
COMP VFD PART:135X1998
REACTOR PART: KDRMA5L1

ELECTRICAL INFORMATION
MOTOR/CTRL CIRCUIT MCA: 29.3A
MOTOR/CTRL CIRCUIT MOP: 30A

RTU VFD SCHEMATIC

COMPONENT LIST

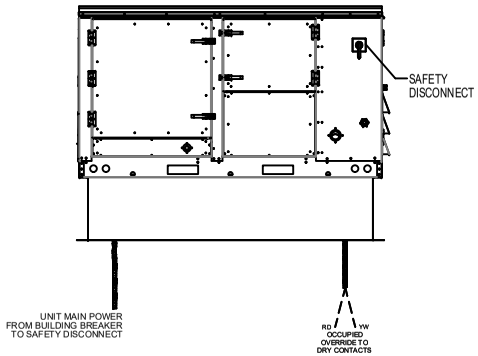
LABEL	DESCRIPTION
MT-01	Supply Motor
PWS-01	24VDC Power Supply
SW-01	Main Disconnect Sw
VFD-1	Supply Motor VFD
VFD-2	Compressor VFD

LEGEND
FIELD WIRING
FACTORY WIRING

BK- BLACK	YW- YELLOW
BL- BLUE	GY- GREY
BR- BROWN	PR- PURPLE
OR- ORANGE	RD- RED
WH- WHITE	GR- GREEN
OR/BL- OR/BL STRIPE	
BL/RD- BL/RD STRIPE	
RD/GN- RD/GN STRIPE	
WH/BL- WH/BL STRIPE	

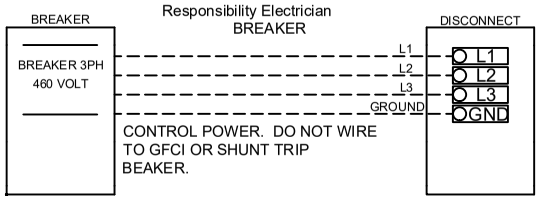
JOB NAME	2102482 - McCom...
DRAWING TITLE	
DESCRIPTION OF OPERATION	MUA BOARD (DETAILS TO FOLLOW)
JOB NO	8252713
DRAWN BY	AUTO
TYPE	FACTORY
DATE	10/14/2025
DWG NO	RTUVFD8252713-1

RTU Installation Wiring	JOB	*2102482 - McComb, MS*
DRAWING NUMBER *INST8252713-1*		MODEL CAS-HVAC1-I.125-13-10T-DOAS

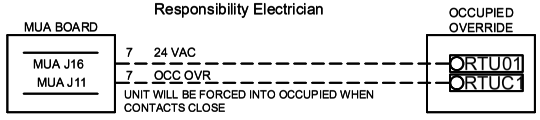


HMI SCHEDULE				
UNIT NUMBER	HMI #	HMI LOCATION	TEMP AVERAGING	MODBUS ADDRESS
FAN #1	HMI #1 - UNIT	IN UNIT	NOT AVERAGED	55
FAN #1	HMI #2 - SPACE		AVERAGED	56

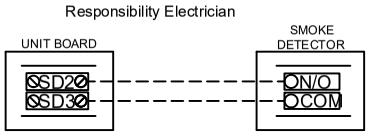
BREAKER TO PRIMARY DISCONNECT



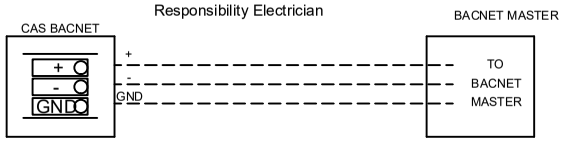
MUA BOARD TO OCCUPIED OVERRIDE



UNIT BOARD TO SMOKE DETECTOR



CAS BACNET TO BACNET MASTER



Installed Options

OCCUPIED OVERRIDE





NOTES

— — — Field Wiring

WIRE COLOR

BK - BLACK	YW - YELLOW
BL - BLUE	GR - GREEN
BR - BROWN	GY - GRAY
OR - ORANGE	PR - PURPLE
RD - RED	PK - PINK
WH - WHITE	

Rigging

WARNING!

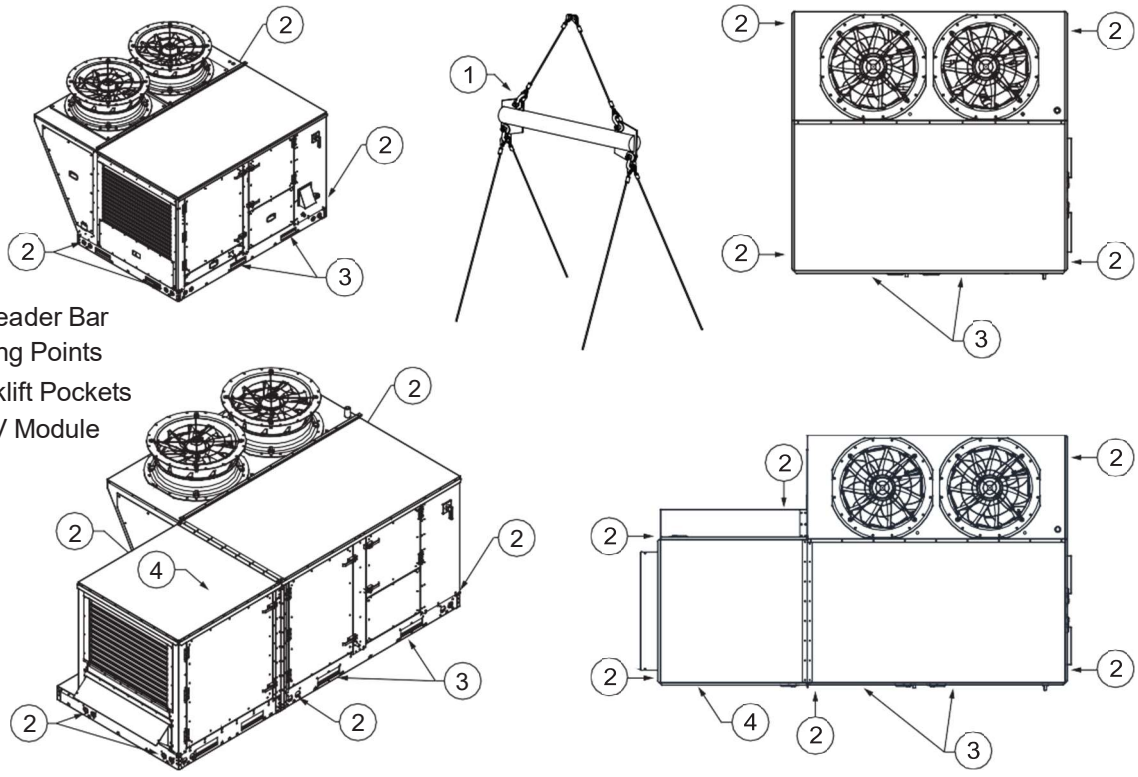
Ensure that all the lifting equipment used is properly rated for the weight of the unit being lifted. Each of the lifting implements (chains, slings, or straps), hooks, and shackles used to lift the unit must be capable of supporting the entire weight of the unit. Each lifting point must use individual lifting implements. Lifting implements may not be of the same length. Adjust as necessary for even unit lift. Other lifting arrangements could cause equipment or property damage. Failure to follow the instructions above or properly lift the unit could result in dropping the unit. This can result in death or serious injury to the operator/technician.

Spreader bars must be used and should extend past the edges of the equipment to avoid damage to the casing. Not using spreader bars may cause damage to the unit casing. Refer to **Figure 3** for rigging details.

WARNING: DO NOT LIFT UNIT BY THE INTAKE LOUVER, OR DOOR OPENINGS – USE ALL LIFTING POINTS PROVIDED WITH A SPREADER BAR OR SLINGS UNDER THE UNIT – USE CARE NOT TO DAMAGE COILS, SWITCHES OR PROTRUDING SHEET METAL COMPONENTS.

- Units are supplied with multiple lifting points (eyes) on the bottom corners of the structural rails.
- Always use spreader bars to prevent damage to the unit casing.
- Test lift the unit approximately **2 feet** to verify proper center of gravity lift point. To avoid dropping unit, re-position lifting point if unit is not level. Failure to properly lift unit could result in unit dropping and possibly crushing operator/technician, which could result in death or serious injury and possible equipment or property damage.

Figure 3 - Rigging (Size 2 Unit Shown)



1. Spreader Bar
2. Lifting Points
3. Forklift Pockets
4. ERV Module

NOTE: For units without ERV, you must use at least four lifting point locations. For ERV units, at least 6 lifting points are required when raising the unit.